# *Fault Tolerant Carry Save Adders – A NMR Configuration Approach*

S.Radhakrishnan, T.Nirmalraj
*Department of ECE, Srinivasa Ramanujan Centre*
*SASTRA Deemed University*
Kumbakonam, India
krishnan2001@gmail.com
nirmalraj.harsha@src.sastra.edu

S.Ashwin, V.Elamaran, Rakesh Kumar Karn
*Department of ECE, School of EEE*
*SASTRA Deemed University*
Thanjavur, India
ashwinsuresh2496@gmail.com
elamaran@ece.sastra.edu

***Abstract*— In digital systems design, the high-speed adders play an imperative role in modules like adders, multipliers, division, etc. to obtain the results quickly. In digital signal processor and computer data path circuits, carry look-ahead adder (CLA), carry save adder (CSA), carry select adder (CSeA) and carry skip adder are the mostly used fast adders to improve the performance of a system. This work mainly emphasizes on fault-tolerant carry save adder using hardware redundancy configurations for the betterment of reliability at the cost of area. The popular methods like dual (DMR) and triple modular redundancy (TMR) are implemented to detect only but not correct an error and tolerate single event upset (SEU) error respectively. To tolerate multiple errors, for example, two, three and four errors, the 5-modular (quintuple), 7-modular (septuple), and 9-modular (nonuple) redundancy configurations are used respectively. For experimental results, the implementation of a simple addition of four 4-bit numbers is considered on Altera FPGA EP4CE115F29C7 device using Quartus II synthesis software tool. Simulation results reveal that for example in nonuple MR, CSA using CSeA final stage obtain higher performance (45.7 MHz) with moderate power dissipation as 173.36 mW and at the cost of more LEs (216).**

***Index Terms*— Carry save adder, FPGA, fault-tolerant, triple modular redundancy, Quartus II***

## I. INTRODUCTION

In the implementation of digital signal processing circuits and microprocessors, arithmetic circuits are indispensable blocks. Even though the complexity of microprocessors and DSP processors increases rapidly, the routine basic operations like an adder, multiplier, divider, shifter, and comparator circuits which are the main components of arithmetic and logic (ALU) remain same [1]. So there is a massive demand in optimizing such circuits regarding area, performance, and power concerning specific applications. The propagation delay is the primary concern for the performance of circuits such as ripple adders, ripple subtractors, and adder/subtractors. The power dissipation too is another major concern for the processing of multi-operand bits [2].

Due to the shrinking dimensions in VLSI technology, faults like permanent, transient and many not modeled faults can occur with the higher rate which results in poor dependability of the system. Safety and mission-critical applications often use redundancy schemes to mitigate any arbitrary function module failure(s)/fault(s) to obtain an uninterrupted and correct operation. So our main goal of this paper is to increase the reliability of the system but at the cost of area due to duplication of circuits [3-7].

The low power high-speed adders are essential for the high-performance units. Since the requirement of the high-performance units grows, there is a tremendous need to enhance the performance of the data path units. An adder with low power, low area, and high performance plays an essential part in majority portion of the hardware circuits. The CLA and CSA architectures have been introduced mainly to reduce the carry propagation delay. Due to the parallel structure in CSA, the carry propagation is alleviated is the main advantage [8].

The main objective of this work is to detect or correct faults in a circuit and hence obtain the fault-free response of the system for the betterment of reliability. The hardware redundancy is the most common technique used to design a fault-tolerant system. A triple modular redundancy (TMR) is the one in which two duplicate copies of the original module are placed along with the actual module; if one module fails or fault occurs in a module, the error-free response can be obtained by computing the majority among all the modules as in Fig. 1. This method is the simple and cost-effective technique to mitigate soft errors in critical applications due to the duplication of modules. This TMR configuration can tolerate one single event upset (SEU) error [9-14].

Let $R_A$, $R_B$, and $_{RC}$ are the reliabilities of modules $A$, $B$, and $C$ respectively to evaluate the reliability of a TMR configuration with a perfect voter. If all the modules function correctly, then the reliability of a TMR system is "$R_A R_B R_C$". If the module $A$ fails, it will become "$(1–R_A)R_B R_C$"; if the module $B$ fails it will become "$R_A(1–R_B)R_C$" and if the module $C$ fails, it will become "$R_A R_B(1–R_C)$". These events are mutually exclusive, and hence the overall TMR system reliability can be expressed as:

$R_{TMR}$ = Probability of no failures + Probability of failure in module $A$ + Probability of failure in module $B$ + Probability of failure in module $C$

$= R_A R_B R_C + (1–R_A)R_B R_C + R_A(1–R_B)R_C + R_A R_B (1–R_C)$

$= R^3 + 3(1–R)\, R^2$

$= 3R^2 – 2R^3$ (if $R_A = R_B = R_C = R$) (1)


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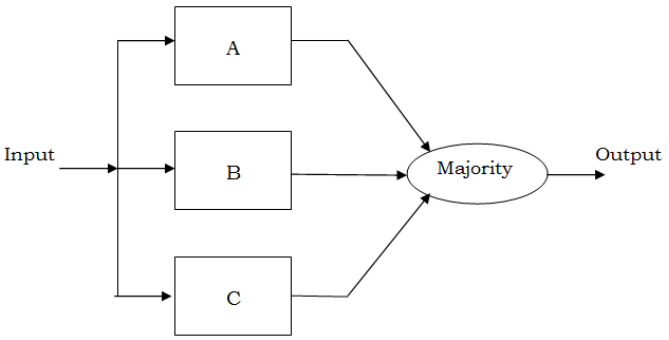



Fig.1. A TMR configuration with a voter.

If the voter is not a perfect one, then the reliability of a voter ($R_V$) will be multiplied and the expression of $R_{TMR}$ will become:

$$R_{TMR} = (3R^2 - 2R^3)\, R_V \quad (2)$$

The majority function of a TMR configuration, i.e. the voter output is expressed as [9,13]:

$$V_{3MR} = AB + BC + CA \quad (3)$$

N-modular redundancy (NMR) can also be used to mitigate SEU errors to improve the reliability, especially to tolerate multiple errors. The 5MR (N=5) system can tolerate two SEU errors which is shown in Fig. 2; the 7MR system can tolerate three SEU errors; the 9MR system can tolerate 4 SEU errors, etc. In general, any NMR system can mask (N–1)/2 SEU errors and hence these configurations help improve the reliability of a system [13].

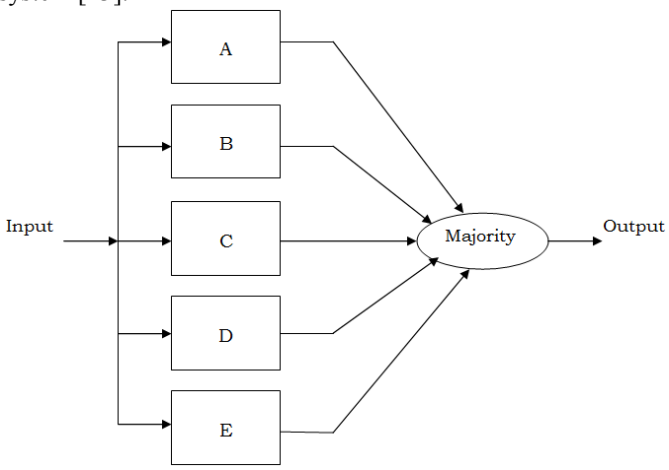



Fig.2. A 5MR configuration with a voter.

Let $R_A$, $R_B$, $R_C$, $R_D$, and $R_E$ are the reliabilities of modules $A$, $B$, $C$, $D$, and $E$ respectively in order to evaluate the reliability of a 5MR configuration with a perfect voter. This is also referred to as quintuple modular redundancy (QunitupleMR). If all the modules function correctly, then the reliability of a TMR system is "$R_AR_BR_CR_DR_E$". If the modules $A$ and $B$ fail, they will become "$(1–R_A)(1–R_B)R_CR_DR_E$"; if the modules $A$ and $C$ fail it will become "$(1–R_A)R_B(1–R_C)R_DR_E$" and so on. Again these events are mutually exclusive and hence the overall 5MR system reliability can be expressed as:

$R_{5MR}$ = Probability of no failures + Probability of failures in module A and B + Probability of failures in module A and C + Probability of failures in module A and D + . . . + Probability of failures in module D and E

$$= R_AR_BR_CR_DR_E + (1–R_A)(1–R_B)R_CR_DR_E + (1–R_A)R_B(1–R_C)R_D\,R_E + \ldots + R_AR_BR_C(1–R_D)(1–R_E)$$
$$= 10R^3(1–R)^2 + 5R^4(1–R) + R^5 \quad (4)$$
(if $R_A = R_B = R_C = R_D = R_E = R$)

If the voter is not a perfect one, then the reliability of a voter ($R_V$) will be multiplied and the expression of $R_{5MR}$ becomes:

$$R_{5MR} = (10R^3(1–R)^2 + 5R^4(1–R) + R^5)\, R_V \quad (5)$$

The majority function of a 5-MR configuration, i.e. the voter output is expressed as [4,8]:

$$V_{5MR} = ABC + ABD + ABE + ACD + ACE + ADE + BCD + BCE + BDE + CDE \quad (6)$$

Similarly, the reliability of a 7-MR system with a perfect voter is expressed as (Fig. 3):

$$R_{7MR} = 35R^4(1–R)^3 + 21R^5(1–R)^2 + 7R^6(1–R) + R^7 \quad (7)$$

The majority function of a 7-MR configuration, i.e. the voter output is expressed as [9,13]:

$$V_{7MR} = ABCD + ABCE + ABCF + ABCG + ABDE + ABDF + ABDG + ABEF + ABEG + ACDE + ACDF + ACDG + ADEF + ADEG + AEFG + BCDE + BCDF + BCDG + BCEF + BCEG + BCFG + BDEF + BDEG + BEFG + CDEF + CDFG + CEFG + DEFG \quad (8)$$

This is also referred to as septuple modular redundancy (SeptupleMR). Similarly, the majority voter of a 9-MR configuration can be obtained. This is also referred to as nonuple modular redundancy (NonupleMR).

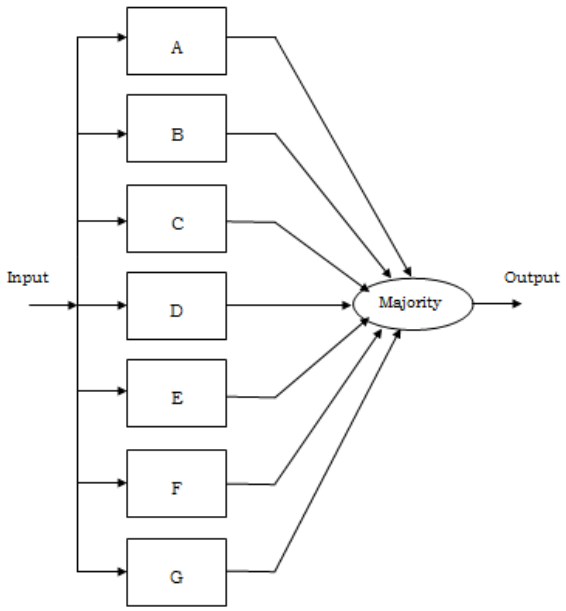



Fig.3. A 7MR configuration with a voter.

This paper is organized as follows: Section II contains the implementation design details of carry save adder. The fault-tolerant CSA circuits using hardware redundancy configurations are described in Section III. Simulation results and discussion are made in Section IV. Finally, Section V describes the conclusion with future work.


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## II. Carry Save Adder Designs

The addition of more than two numbers is done using carry-save adders using full-adders as basic building blocks. The full adder has three inputs and two outputs which becomes the principle behind in designing the carry-save adder. The addition of four 4-bit numbers using CSA is shown in Figure 4. Here, the final sum is obtained using ripple-carry adder (RCA). This work demonstrates the multi-operand addition using CSA with different final stages like carry look-ahead adder (CLA) and carry select adder (CSeA) along with the RCA stage [15-16].

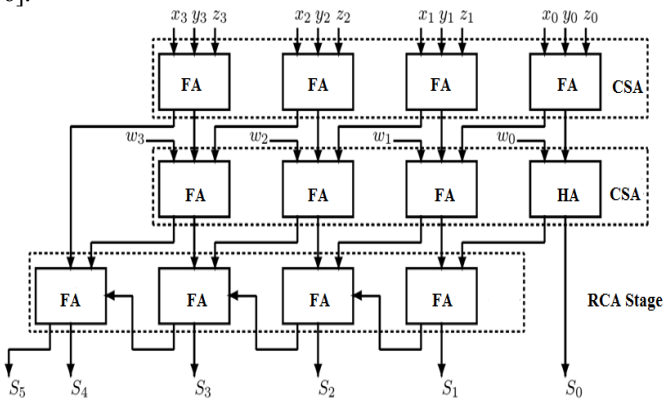



Fig.4. Addition of four 4-bit numbers.

### A. The final stage of CSA using RCA

A four-bit ripple carry adder used as a final stage to compute addition of four 4-bit numbers is shown in Figure 5. Each full adder will compute carry out only if the carry in is available which creates more delay to the circuit [9]. The path of the worst case delay through this adder is: cin – c1 – c2 – c3 – carry. For simplicity, only the partial VHDL script of a 4-bit RCA is given in Figure 6.

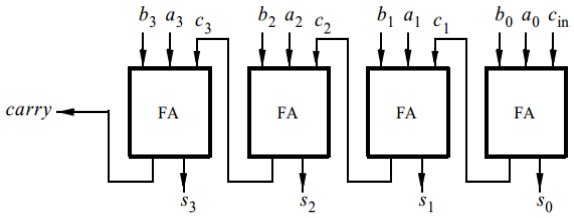



Fig.5. A 4-bit RCA.

```
row3_0: fulladder port map(cin1,m8,m9,s1,m15);
row3_1: fulladder port map(m15,m10,m11,s2,m16);
row3_2: fulladder port map(m16,m12,m13,s3,m17);
row3_3: fulladder port map(m17,m14,m7,s4,cout);
```

Fig.6. A portion of the VHDL script of a 4-bit RCA.

### B. The final stage of CSA using CLA

The Carry Look-Ahead adders (CLA) play a huge role in producing high speed in the design of digital arithmetic circuits based on the expression,

$$C_{i+1} = (A_i \oplus B_i)C_i + A_i B_i \quad (9)$$

The schematic view of this expression is shown in Figure 7. Since the carry is being evaluated based on the "generate (AB)" and "propagate (A B)" terms, the delay is much reduced as compared to the RCA circuit. The schematic of the generation of carry bits for a 4-bit CLA is shown in Figure 8. The VHDL script of propagate, generate, carry, and sum bits a 4-bit CLA is given in Figure 9 [16].

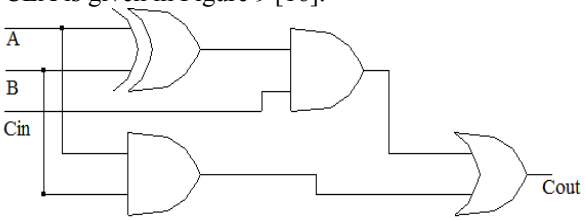



Fig.7. The carry-out schematic using CLA concept.

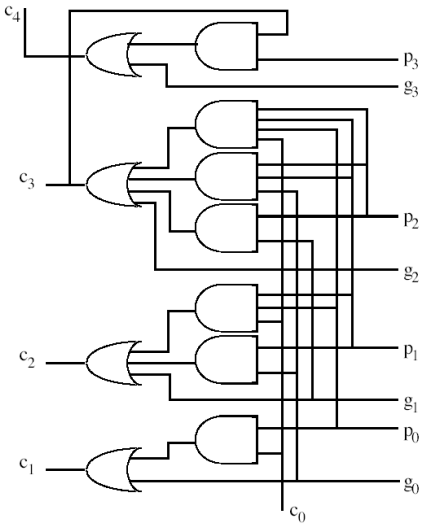



Fig.8. Carry bits generation for a 4-bit CLA.

```
p0 <= m8 xor m9; g0 <= m8 and m9; c11 <= (p0 and cin1) or g0; s1 <= p0 xor cin1;
p1 <= m10 xor m11; g1 <= m10 and m11; c21 <= (p1 and c11) or g1; s2 <= p1 xor c11;
p2 <= m12 xor m13; g2 <= m12 and m13; c31 <= (p2 and c21) or g2; s3 <= p2 xor c21;
p3 <= m7 xor m14; g3 <= m7 and m14; cout <= (p3 and c31) or g3; s4 <= p3 xor c31;
```

Fig.9. A portion of the VHDL script of a 4-bit RCA using CLA concept.

### C. The final stage of CSA using CSeA

The carry-select adder (CSeA) is used in the final stage of a CSA to add four 4-bit numbers [14]. This is used especially when adding more number of bits in the CSA tree structure. Figure 10 shows the VHDL script of this module partially.

## III. Fault-Tolerant CSA with Hardware Redundancy Configurations

In the introduction section, all kinds of redundancy configurations are discussed in detail with the reliability expressions except dual and quadruple modular redundancy configurations.


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```
zero <= '0'; one <= '1';
row3_0: fulladder port map(cin1,m8,m9,s1,m15);
row3_1: fulladder port map(m15,m10,m11,s2,m16);
row3_2: fulladder port map(zero,m13,m12,s30,m170);
row3_3: fulladder port map(m170,m7,m14,s40,cout0);
row3_4: fulladder port map(one,m13,m12,s31,m171);
row3_5: fulladder port map(m171,m7,m14,s41,cout1);
s3 <= ((not(m16)) and s30) or (m16 and s31);
s4 <= ((not(m16)) and s40) or (m16 and s41);
cout <= ((not(m16)) and cout0) or (m16 and cout1);
```

Fig.10. Partial VHDL script of a 4-bit RCA using final stage with CSeA.

### A. Dual Modular Redundancy (DMR)

This configuration is often used for error detection only. If one module fails, due to the mismatch, this will convey the error indication as in Figure 11. This scheme will not correct the error, and hence the configuration is simple. Figure 12 shows the VHDL script portion of the error detection using DMR configuration.

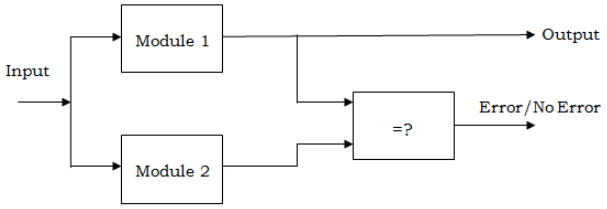



Fig.11. DMR configuration with error detection.

```
s0 <= s01; s1 <= s11; s2 <= s21; s3 <= s31; s4 <= s41; cout <= cout1;
error1 <= s01 xor s02; error2 <= s11 xor s12; error3 <= s21 xor s22;
error4 <= s31 xor s32; error5 <= s41 xor s42; error6 <= cout1 xor cout2;
error <= error1 or error2 or error3 or error4 or error5 or error6;
```

Fig.12. Partial VHDL script of error detection using DMR.

### B. Quadruple Modular Redundancy (QMR)

The output of this QuadMR is high if a majority of the inputs are high as shown in Figure 13. The Boolean expression for this voter is expressed as,

$$Z = A0\,A1\,A2\,\overline{A3} + A0\,A1\,\overline{A2}\,A3 + A0\,\overline{A1}\,A2\,A3 + A1\,A2\,A3 \quad (10)$$

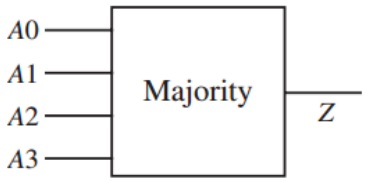



Fig.13. Majority circuit with QuadMR configuration.

## IV. Simulation Results and Discussion

The addition of four 4-bit numbers is implemented on Altera FPGA EP4CE115F29C7 device using Quartus II synthesis tool with hardware redundancy configurations. Figure 14 shows the response of the addition of four 4-bit numbers with fault injection using TMR configuration. The number of logic elements (LEs), power dissipation and performance results are obtained and compared in this section.

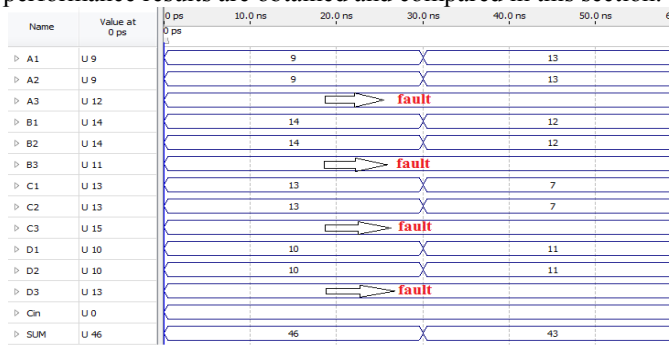



Fig.14. Addition of four 4-bit numbers with TMR configuration.

### A. Logic Elements

Figure 15 shows the number of logic elements (LEs) resources used in FPGA device and the results reveal that the CSA-RCA and CSA-CLA design styles offer lower LEs than the CSA-CSeA style. For example, in the Nonuple MR configuration, CSA-RCA and CSA-CLA design styles needs 336 LEs while the CSA-CSeA style need 344 LEs.

### B. Performance

Figure 16 shows the comparison of performance results, and this reveals that the CSA-CSeA combination offers better performance than others. For example, in the Septuple MR configuration, CSA-CSeA style provides better performance (45.7 MHz) than CSA-RCA (44.5 MHz) and CSA-CLA (38.6 MHz).

### C. Power Dissipation

The power dissipation results of all the styles are shown in Figure 17. It is evident that the CSA-CSeA combination requires lower power dissipation than others. For example, in the Qunituple MR configuration, CSA-CSeA style requires lower power dissipation (134.40 mW) than CSA-RCA (142.67 mW) and CSA-CLA (144.63 mW).

## V. Conclusion

The demand for fault-tolerant system designs grows abundantly in the era of microelectronics circuit design. This study exemplifies the fault-tolerant carry save adder using hardware redundancy configurations to tolerate one, two, three, and four SEU bit errors using TMR, 5-MR, 7-MR and 9-MR configurations respectively. The different styles like CSA-RCA, CSA-CLA, and CSA-CSeA are demonstrated to add four 4-bit numbers, and the results are analyzed with an Altera FPGA EP4CE115F29C7 device using Quartus II synthesis tool. This work can further be extended to other high-speed adders like carry-ripple adders, carry-skip adders, carry look-ahead adders and carry-select adders too. Also, this study would help to design fault-tolerant high-speed multipliers which will make a high impact in the era of VLSI design.


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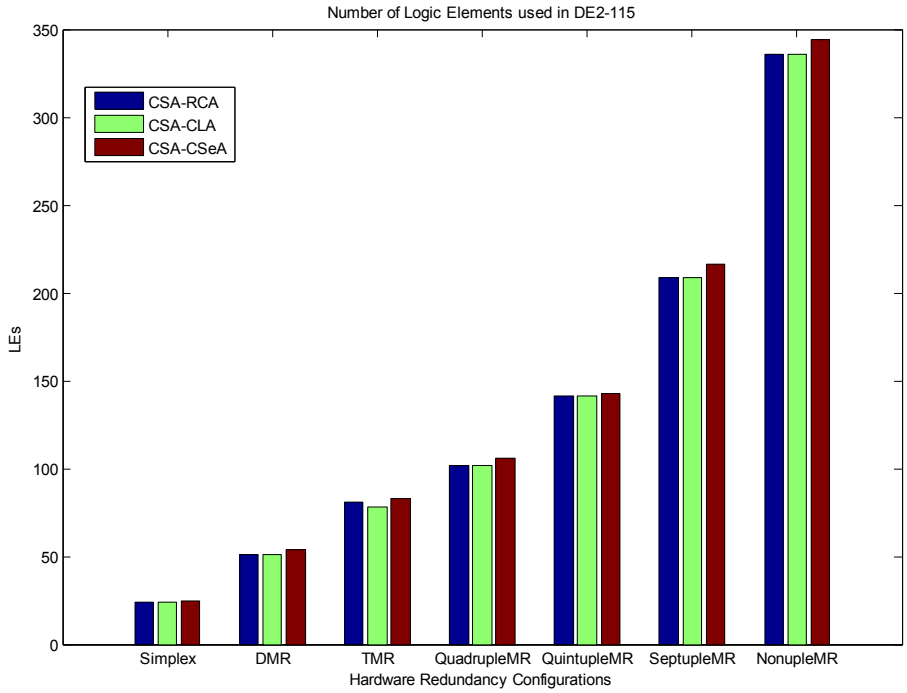



Fig.15. The usage of logic elements.

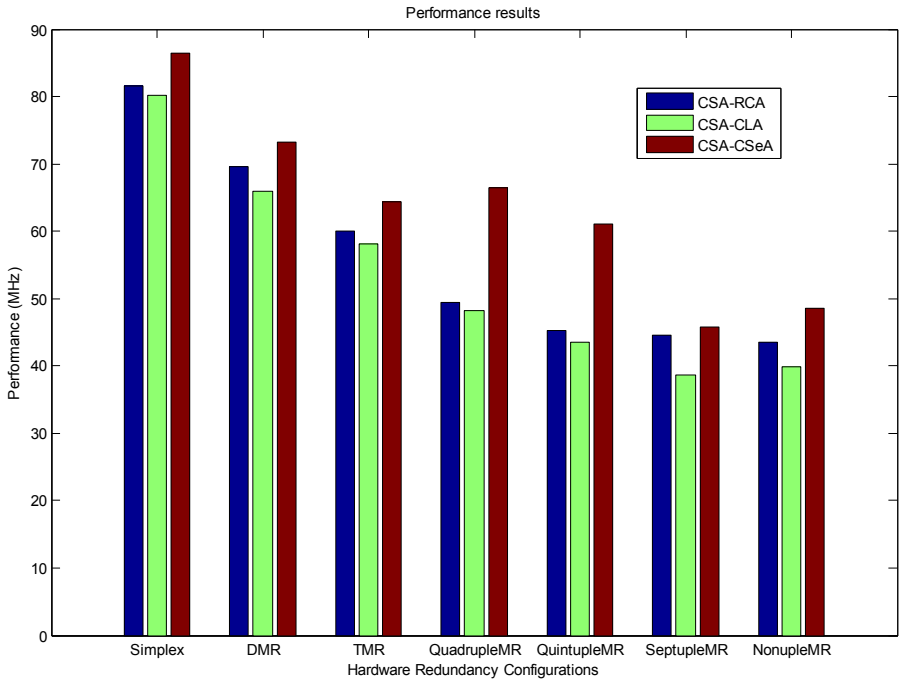



Fig.16. Performance results.

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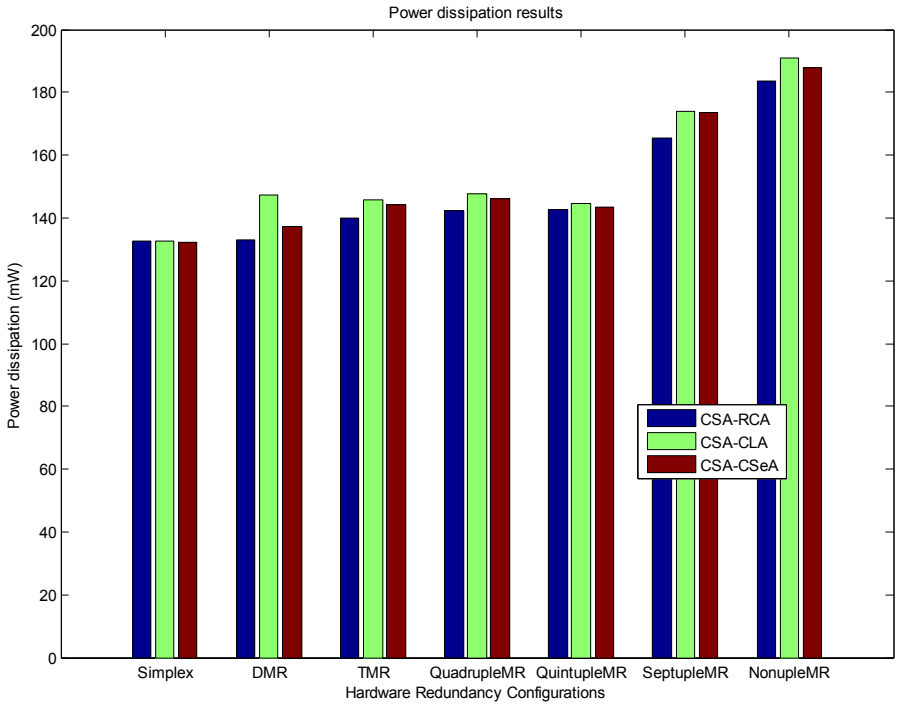



Fig.17. Power dissipation results.

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